

Applying Kalman Filter in EEG-Based Brain Computer Interface for Motor Imagery Classification

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Abstract—Human imagination and intention can be read by using the Electroencephalography(EEG)-Based Brain Computer Interface(BCI) for Motor Imagery. The systems reads the human brain which consists of left or right imaginary movement and decide the actual movement. It is useful especially for the disable people to help their daily life. But the signals include the unwanted signals and other features together with the actual signal. The challenges are to remove the unwanted signals and extract the accurate feature and classify the signal accurately by using the best classification method. This paper applied the Kalman filter to the BCI's signal processing methods to improve the accuracy and the reliability of the systems. The Common Spatial Pattern (CSP) is used to extract the feature and the Radial Basis Function (RBF) as the classification method. We also compared other classification method which is Fisher Linear Discriminant Analysis (FLDA) to show the RBF gives better result. The simulation result shows improvement by using the proposed method compared to the other method.

I. INTRODUCTION

Brain Computer Interface or BCI is useful as a communication pathway between the human and other devices. The BCI can read the human intention by analyzing the human brain's signal from EEG (Electroencephalography). Then it can be expanded by sending commands to some devices. So it is useful in order to help disabled people in their daily life. Simply by thinking, they can tell others their intention and able to communicate with assistance devices particularly for those who cannot use a keyboard or mouse.

EEG is the dominant technique used in brain measurement due to its cost effectiveness and easy implementation. The electrodes take only several minutes to read the brain signal from the scalp. Thus, there is no sensor needed to be implanted inside the scalp. However, the signals measured are affected by the noise which will reduce the accuracy of the BCI system in recognizing the real intention. Some signal can be very bad depends on the unwanted signal occurrence. Therefore, the signal processing in BCI system required several stages consist of pre-processing the signal, signal feature extraction and signal classification.

The studies on signal processing on the BCI systems has been developed for years. In 2005 [1], Common Spatial Pattern (CSP) is used for channel selection. The optimal

channel is selected based on maximum vector derived from CSP.

The complex Morlet wavelet transform has been proposed in 2008 [2]. This method used both alpha and beta rhythms. It detected and classified the left and right movement imagination. The Feature Subset Selection algorithm is used as the feature extraction and the Multilayer Backpropagation Neural Network as the classification.

In 2011 [3], the Regularizing Common Spatial Pattern has been introduced in replace to the normal CSP. The accuracy has been improved for some dataset but not much improvement in bad signal dataset.

In this paper, the Kalman filter has been applied to the signal processing of the BCI system. The system used CSP as the feature extraction and RBF as the classification and some improvement can be seen in terms of accuracy of the system. The goal of this study is to improve the accuracy of the motor imagery classification especially the signal with a lot of noise.

II. SYSTEM MODEL

A. Pre-processing

When EEG data is recorded, it also recorded some unwanted signal such as eyes movement or muscle movement. So Pre-processing is used to filtered out the unwanted signal. For each trial, the lowpass filtered in 0.5Hz to 30Hz is using fifth order Butterworth filter.

B. Common Spatial Pattern

Common Spatial Pattern, CSP is used in signal processing to separate multichannel signal from two classes to additive subcomponents. It is used by maximizing the variances of two-class signal matrices. The idea is to get the largest and smallest variance of each classes [1] which is called as CSP weight. The weight is used to get the projected signal and eliminates the CZ value from the training data.

C. Radial Basis Function

Radial Basis Function, RBF is used as a function classification and mean square error minimization on training data [4]. It calculates the distance of the point of the input to the origin. In order to find the weight, W of of the basis function

the coefficient of basis function, α is divided by the training output, Y (1). The basis function can be chosen either to be Gaussian, Cubic, Thin Plate Spline Cauchy, Multiquadric, Inverse Multiquadric or Identity. Here the function we used taken by Gaussian and the coefficient can be calculated by using (2) equation. The output can be formed by applied the weight of the basis function to the distance. The output will be used to get the decision function to classify the signals means of classifier.

$$W = \alpha/Y(1)$$

$$\alpha(||x - c_i||) = \exp[-\beta||x - c_i||^2](2)$$

D. Fisher Linear Discriminant Analysis

Another method to classified the classes in the BCI is by using the Fisher Linear Discriminant Analysis or FLDA. It is used to classify between two classes with simple computation requirements. The weight vector and the threshold will be used to calculate the decision function, $g(v)$ (3) which will be used to classify the class. Where V is the feature vector of the training data.

$$g(v) = a_0 + a_1 \times v(3)$$

E. Kalman filter

We applied Kalman filter to improve the accuracy of the classification output by get the correct estimation [5]. By using the Kalman filter, the CSP weight, W_{csp} is updated to get the best projected signal and accuracy. The W_{csp} is updated by introduce the error to the system. There are two noises introduce to the system. The measurement noise comes from the muscle movement and eye blink while the process noise comes from the distraction from the environment. The W_{csp} for the current state depends on the value of the previous state and the process noise, w (4). In (5) the measurement value is calculated by adding the measurement noise, v to the signal value, W_{csp} . The measurement noise is assumed to be random Gaussian noise. The process noise and measurement noise are statistically independent.

$$W_{csp_k} = fW_{csp_{k-1}} + w_{k-1}(4)$$

$$y_k = HW_{csp_k} + v_k(5)$$

III. SIMULATION

A. Dataset

The dataset used in the simulation is from the BCI Competition II Dataset III. The dataset was from EEG signal of a normal 25 years old female subject. The EEG recorded the motor imagery signal for right hand movement and left hand movement. The electrode arrangement used the 10-20 systems to measure the signal from 3 channels which are C3, CZ and C4. During the experiment, each trial took 9 seconds with 1152 samples data. There is 140 trials as the training data and another 140 trials for the test data and the actual data which record the actual classes for each trial. Figure

1 shows the timing scheme during the experiment. During the first 2 seconds, nothing happened and was quiet. At 2 second the beep sound as the trigger can be heard and the visual cue will be appeared. The motor imagery will started at 3 second until 9 second. During this time, the arrow that shows either left or right will appeared.

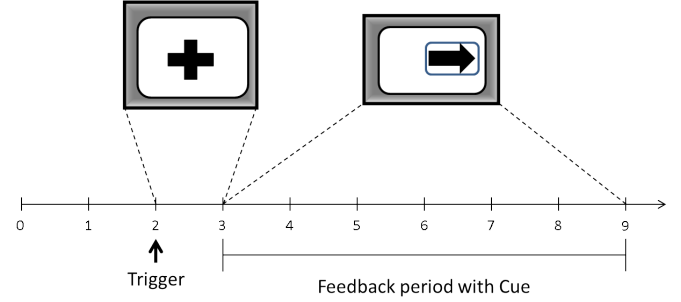


Figure 1: Timing scheme during the experiment

B. Result

We used several methods to compare the result and it shows the improvement in the proposed method. The methods are, FLDA without feature extraction, CSP as Feature Extraction and FLDA as Classification method, Kalman filter applied to the CSP and FLDA method and CSP as feature extraction with RBF as classification method and the proposed method which Kalman filter method applied to the CSP as the feature extraction and RBF as classification method. Figure 2 shows the results for all the methods. The proposed method gives the best accuracy compared to others as the W_{csp} is corrected and projected the better signal to the systems by using Kalman filter.

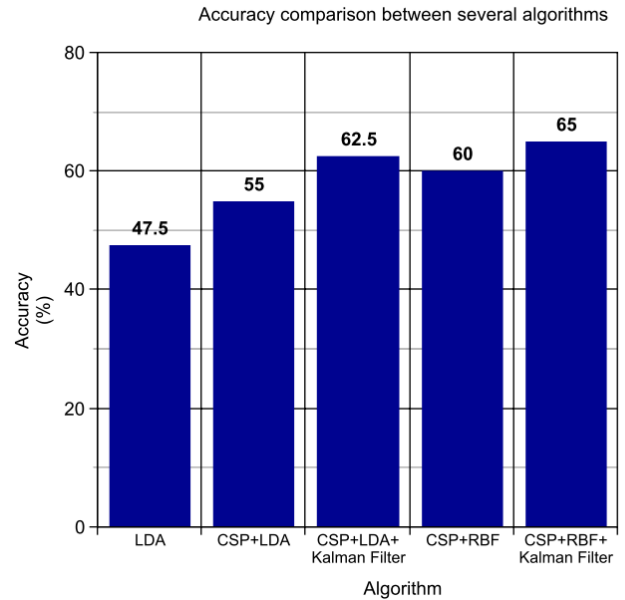


Figure 2: Accuracy comparison between several algorithms

Compared to the FLDA, the RBF gives the better result in

term of accuracy as RBF minimizes the mean square error of the training data and maximizes the distance between the training data and the decision function. So RBF is able to determine better which side of the class the feature vector belongs to.

While applying Kalman filter, both method showed better results than the method without it.

IV. CONCLUSION AND FUTURE WORKS

In summary, we applied the Kalman filter to the Brain Computer Interface(BCI) to the combination of Common Spatial Pattern(CSP) as the feature extraction and Radial Basis Function (RBF) as the classification. The classification of the motor imagery for the right and left hand movement is improved in terms of accuracy compared to other methods. The improvement is important to make the system and the signal data more reliable.

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